6.1 2022204065 윤성근

1

(1) $f(t) = \sin 2t$

$L(f(t)) = L(\sin 2t)$

$= \frac{2}{s^2+4}$ //

2

(1) $\frac{1}{s^5}$

$L^{-1}\left(\frac{1}{s^5}\right) = t^4 \cdot \frac{1}{120}$ //

↓ $L(t^n) = \frac{n!}{s^{n+1}}$

6.2

1

(1) $e^{-t}\cosh t$

$L(\cosh t) = \frac{s}{s^2-1}$,

$L(e^{-t}\cosh t) = \frac{s}{s^2-1}\Big|_{s\to s+1} = \frac{s+1}{(s+1)^2-1}$

$= \frac{s+1}{s^2+2s}$ //

2

(1) $\frac{(s+1)^3}{s^4}$

$\frac{(s+1)^3}{s^4} = \frac{s^3+3s^2+3s+1}{s^4} = \frac{1}{s} + \frac{3}{s^2} + \frac{3}{s^3} + \frac{1}{s^4}$

$L^{-1}\left(\frac{1}{s}\right) = 1$

$L^{-1}\left(\frac{3}{s^2}\right) = t^1 \cdot 3 = 3t$

$L^{-1}\left(\frac{3}{s^3}\right) = t^2 \cdot \frac{3}{2} = \frac{3}{2}t^2$

$L^{-1}\left(\frac{1}{s^4}\right) = t^3 \cdot \frac{1}{3!} = \frac{1}{6}t^3$ //

$\therefore L^{-1}\left(\frac{(s+1)^3}{s^4}\right)$

$= \frac{1}{6}t^3 + \frac{3}{2}t^2 + 3t + 1$ //

4

(1) $y' - 3y = \sin 2t$, $y(0) = 2$

$L(y') - L(3y) = L(\sin 2t)$

$(L(y(t)) = Y(s))$

$sY(s) - y(0) - 3Y(s) = \frac{2}{s^2+4}$

$(s-3)Y(s) = \frac{2}{s^2+4} + 2$

$Y(s) = \frac{2}{(s^2+4)(s-3)} + \frac{2}{s-3}$

$L^{-1}(Y(s)) = y(t)$

$L^{-1}\left(\frac{2}{(s^2+4)(s-3)} + \frac{2}{s-3}\right)$

$= L^{-1}\left(\frac{2}{(s^2+4)(s-3)}\right) + L^{-1}\left(\frac{2}{s-3}\right)$

$= L^{-1}\left(\frac{\alpha s+\beta}{s^2+4} + \frac{k}{s-3}\right)$

$\frac{(\alpha s+\beta)(s-3)+k(s^2+4)}{(s^2+4)(s-3)}$

$\alpha s^2 - 3\alpha s + \beta s - 3\beta + ks^2 + 4k = 2$

$\begin{cases} \alpha + k = 0 \\ -3\alpha + \beta = 0 \\ -3\beta + 4k = 2 \end{cases}$

$3k + \beta = 0$

$13k = 2$

$k = 2/13$, $\alpha = -2/13$, $\beta = -6/13$

$$L^{-1}\left(\frac{-\frac{2}{13}s}{s^2+4} + \frac{-\frac{6}{13}}{s^2+4} + \frac{\frac{2}{13}}{s-3} \right)$$

$$= -\frac{2}{13}\cos 2t - \frac{3}{13}\sin 2t + \frac{2}{13}e^{3t} \quad //$$

$$\therefore y(t) = -2/13 \cos 2t - 3/13 \sin 2t + 2/13 \cdot e^{3t} \quad //$$

6.3

#1

(2) $f(t) \begin{cases} \sin t & (0 \le t < 2\pi) \\ 0 & (t \ge 2\pi) \end{cases}$

$$f(t) = \sin t + U(t-2\pi)(-\sin t)$$

$$L(f(t)) = L(\sin t) + L(-\sin t\, U(t-2\pi))$$

$$= \frac{1}{s^2+1} + L(-\sin(t-2\pi)U(t-2\pi))$$

$$= \frac{1}{s^2+1} + e^{-2\pi s} L(-\sin t)$$

$$= \frac{1}{s^2+1} - e^{-2\pi s} \cdot \frac{1}{s^2+1}$$

$$= \frac{1-e^{-2\pi s}}{s^2+1} \qquad L^{-1}\left(\frac{1-e^{-2\pi s}}{s^2+1}\right) = f(t)$$

//

#2.

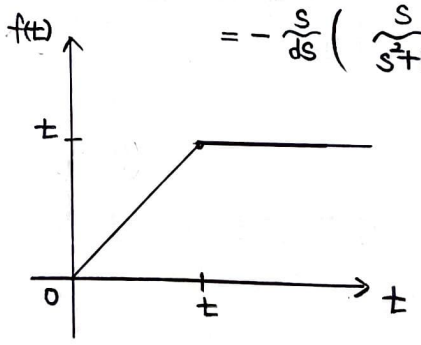
(1) $f(t) = L^{-1}\left(\frac{1}{s^2} - \frac{e^{-s}}{s^2} \right)$

$$L^{-1}\left(\frac{1}{s^2}\right) = t$$

$$L^{-1}\left(\frac{e^{-s}}{s^2}\right) = (t-1)\,U(t-1)$$

$$\therefore f(t) = t - (t-1)U(t-1)$$

$$= \begin{cases} t & (0 \le t < 1) \\ 1 & (t \ge 1) \end{cases}$$



#5

(2) $y'' + 16y = \delta(t-2\pi)$, $y(0)=0$, $y'(0)=0$

$$L(y'') + 16L(y) = L(\delta(t-2\pi))$$

$$s^2Y(s) - s\cancel{y(0)} - \cancel{y'(0)} + 16Y(s) = e^{-2\pi s}$$

$$(s^2+16)Y(s) = e^{-2\pi s}$$

$$Y(s) = \frac{e^{-2\pi s}}{s^2+16} \quad //$$

$$L^{-1}(Y(s)) = f(t)$$

$$L^{-1}\left(\frac{1}{s^2+16} \cdot \frac{1}{4} \cdot e^{-2\pi s} \right)$$

$$= \frac{1}{4}\sin 4t\, U(t-2\pi) \quad //$$

$$= \frac{1}{4}\sin 4(t-2\pi)\, U(t-2\pi) \quad //$$

6.4

#1 $F(s)G(s) = L\left(\int_0^t f(u)\,g(t-u)\,du\right)$

(1) $L\left(t \cdot \int_0^t \sin u\,du\right)$

$$\int_0^t \sin u\,du = -\cos t + 1$$

$$L(t - t\cos t)$$

$$= L(t) - L(t\cos t)$$

$$L(t\cos t)$$

$$= -\frac{s}{ds}\left(\frac{s}{s^2+1}\right) = \frac{-(s^2+1-2s^2)}{(s^2+1)^2}$$

$$= \frac{s^2-1}{(s^2+1)^2}$$

$$\therefore L\left(t\int_0^t \sin u\,du\right)$$

$$\frac{(s^2+1)^2 - s^2(s^2-1)}{s^2(s^2+1)^2} = \frac{1}{s^2} - \frac{s^2-1}{(s^2+1)^2}$$

$$= \frac{3s^2+1}{s^2(s^2+1)^2} \quad //$$

2

(1)

$$f(t) = 3t^2 - e^{-t} - \int_0^t f(u)\, e^{t-u}\, du$$

$$L(f(t)) = L(3t^2) - L(e^{-t}) - L\left(\int_0^t f(u)e^{t-u}du\right)$$

$$Y(s) = 3\cdot \frac{2}{s^3} - \frac{1}{s+1} - Y(s)\cdot \frac{1}{s-1}$$

$$\frac{s}{s-1} Y(s) = \frac{6}{s^3} - \frac{1}{s+1}$$

$$Y(s) = \frac{6(s-1)}{s^4} - \frac{s-1}{s(s+1)}$$

$$= \frac{6}{s^3} - \frac{6}{s^4} - \frac{s}{s(s+1)} + \frac{1}{s(s+1)}$$

$$= \frac{6}{s^3} - \frac{6}{s^4} - \frac{1}{s+1} + \frac{1}{s} - \frac{1}{s+1}$$

$$= \frac{6}{s^3} - \frac{6}{s^4} - \frac{2}{s+1} + \frac{1}{s}$$

$\therefore f(t) = 3t^2 - t^3 - 2\cdot \sin t + 1$ //

#2.

(2)

$$y'(t) = \cos t + \int_0^t y(u)\cdot \cos(t-u)\,du;\ y(0)=1$$

$$L(y') = \frac{s}{s^2+1} + L(y)\cdot \frac{s}{s^2+1}$$

$$sY(s) - y(0) = \frac{s}{s^2+1} + Y(s)\cdot \frac{s}{s^2+1}$$

$$Y(s)\left(s - \frac{s}{s^2+1}\right) = \frac{s}{s^2+1} + 1$$

$$Y(s)\cdot \frac{s(s^2+1)-s}{s^2+1} = \frac{s+s^2+1}{s^2+1}$$

$$Y(s) = \frac{s^2+s+1}{s^3}$$

$$= \frac{1}{s} + \frac{1}{s^2} + \frac{1}{s^3}$$

$$f(t) = 1 + t + \frac{1}{2}t^2$$

종합문제

1.

(1) $y'' - 4y' + 4y = t^3,\ y(0)=1,\ y'(0)=0$

$$L(y'') - 4L(y') + 4L(y) = L(t^3)$$

$$s^2Y(s) - sy(0) - y'(0) - 4(sY(s) - y(0)) + 4Y(s) = \frac{6}{s^4}$$

$$s^2Y(s) - s - 4sY(s) + 4 + 4Y(s) = \frac{6}{s^4}$$

$$(s-2)^2 Y(s) = \frac{6}{s^4} + s - 4$$

$$Y(s) = \frac{6}{s^4(s-2)^2} + \frac{s}{(s-2)^2} - \frac{4}{(s-2)^2}$$

$a+e+f=0$

$$= \frac{a}{s} + \frac{b}{s^2} + \frac{c}{s^3} + \frac{d}{s^4} + \frac{e}{s-2} + \frac{fs+g}{(s-2)^2}$$

$$\frac{as^3(s-2)^2 + bs^2(s-2)^2 + c(s-2)^2 + d(s-2)^2 + es^4(s-2) + s^4(fs+g)}{s^4(s-2)^2}$$

$$(s-2)^2(as^3+bs^2+cs+d) + s^4(es-2e+fs+g)$$

$$(e+f)s - 2e + g$$